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**MATH 525**

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**Probability**

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## 0.1 Intro

Recommended Reading: Probability and Random Processes (Grimmett/Stirzker)

Mathematical study of randomness (random  $\leftrightarrow$  deterministic)

### How do we model Randomness?

Coin:  $P(H) = P(T) = 0.5$

Biased Coin:  $P(H) = 0.7, P(T) = 0.3$

Dice:  $P(1) = P(2) = \dots = P(6) = \frac{1}{6}$

$P(1 \text{ or } 3) = P(\{1, 3\}) = P(1) + P(3) = \frac{1}{3}$

Power Law:  $P(n) = \frac{6}{\pi^2} \frac{1}{n^2}$

$P(n \text{ is even}) = P(\{2, 4, 6, \dots\}) = \frac{1}{2}$

$P(4 \text{ divides } n) = P(\{4, 8, \dots\})$

### Sample space and probability measure

**Definition 0.1.1** (Sample Space). Let  $\Omega$  be a set. Let  $F$  be a collection of subsets of  $\Omega$ , which is a  $\sigma$ -algebra

1.  $\emptyset \in F$
2. If  $A \in F$ , then  $A^c \in F$
3. If  $A_1, A_2, \dots \in F$  then  $\bigcup_n A_n \in F$

**Definition 0.1.2** (Probability Measure). A Probability measure on  $(\Omega, F)$  is a function  $\mathbb{P} : F \mapsto [0, 1]$  s.t.

- $\mathbb{P}(\emptyset) = 0, \mathbb{P}(\Omega) = 1$
- If  $A_1, A_2, \dots \in F$  is disjoint collection then  $\mathbb{P}(\bigcup_n A_n) = \sum_n \mathbb{P}(A_n)$

### Exercise 0.1.3.

- $A, B \in F$

*Proof.*  $A \cap B = (A^c \cup B^c)^c$

$A^c, B^c \in F \implies A^c \cup B^c \in F \implies (A^c \cup B^c)^c \in F$

□

- If  $A, B \in \mathcal{F}, A \subset B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$

$$B = A \cup (B \setminus A) \implies \mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \implies \mathbb{P}(A) \leq \mathbb{P}(B)$$

# 1 Events

## Why Events

When three coins are tossed but not seen, we still take the sample space as:

$$\Omega = \{(b_1, b_2, b_3) \mid b_i \in \{H, T\}\}$$

After uncovering the first (or just after the first toss: is there a difference?)

$$F = \{\emptyset, \Omega, \{(HHH, (HTH), (HTT), (HHT))\}, \{(T, b_2, b_3) : b_i \in \{H, T\}\}\}$$

After uncovering the second,

$$\Omega_0 = \{H, T\}$$

$F = \{\emptyset, \Omega, \{H\} \times \{H\} \times \Omega_0\}$  plus other stuff and unions of stuff After uncovering the third,  $F = P(\Omega)$

## A glimpse of continuum

Let us model the height of a 'random' person

Alice uses a ruler in millimeters.

$$\Omega = \{\dots, \{1.7745, 1.775\}, \dots\}$$
 Event is what my rulers says

Bob uses a ruler in micrometers

$$\Omega = \{\dots, \{1.7777745\}, \dots\}$$

God uses a ruler

$$\Omega = \mathbb{R}[0, 3]$$

Don't do this set  $\Omega = \mathbb{R}$  and let  $F = thatstuff$

## Terminology

$\Omega$ : Sample Space = all possible outcomes

$(\Omega, F)$ : Measurable space what you can know about what happened

$(\Omega, F, \mathbb{P})$ : Probability Space how likely it is that an event happens

Note: If  $\Omega$  is finite, one can put the uniform measure, i.e.

$$\forall \omega \in \Omega, \mathbb{P}(\{\omega\}) = \frac{1}{|\Omega|}$$

## Properties

- $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$
- $\mathbb{P}(A) + \mathbb{P}(B) = \mathbb{P}(A \cup B) + \mathbb{P}(A \cap B)$

**Example 1.0.1.** Union bound:  $\mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n)$

$$\bigcup_n A_n = A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus (A_2 \cup A_1)) \cup \dots$$

$$\mathbb{P}(\bigcup_n A_n) = \mathbb{P}(A_1) + \mathbb{P}(A_2 \setminus A_1) + \mathbb{P}(A_3 \setminus (A_2 \cup A_1)) + \dots$$

Since all the set minus is  $\leq \mathbb{P}(A_n)$  thus  $\mathbb{P}(\bigcup_n A_n) \leq \sum_n \mathbb{P}(A_n)$

## Conditional Probability and Independence

**Definition 1.0.2.**  $A, B \in F, \mathbb{P}(B) \neq 0$

The probability of  $A$  (conditionon/given)  $B$  is  $\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$

**Definition 1.0.3.**  $A_1, A_2, \dots \in F$  and finite subcollection  $i_1 < i_2 < \dots < i_n$

$$\mathbb{P}(A_{i_1} \cap A_{i_2} \cap \dots \cap A_{i_n}) = \mathbb{P}(A_{i_1}) \mathbb{P}(A_{i_2}) \dots \mathbb{P}(A_{i_n})$$

( $A$  and  $B$  are independent if  $\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B)$ )

### Example 1.0.4.

Independent weather/biased coin (70% H 30% T) toss.  $\mathbb{P}(\text{rains}) = 70\%$

$$\Omega = \{(\text{rains}, H), (\text{rains}, T), (\text{clear}, H), (\text{clear}, T)\}$$

$$(\text{rains}) = \{(\text{rains}, H), (\text{rains}, T)\}$$

$$(\text{heads}) = \{(\text{rains}, H), (\text{clear}, H)\}$$

$$(\text{rains and heads}) = (\text{rains}) \cap (\text{heads})$$

Soothsayer's count toss: rain if heads, clear if tails

$$\Omega = \{(\text{rains}, H), (\text{clear}, T)\}$$

$$\mathbb{P}(\text{rains}) = 70\%$$

Independent and identically distributed (i.i.d) tosses:

$$n \text{ tosses, } \mathbb{P}(n\text{heads}) = 0.7^n$$

$$\mathbb{P}(atleast1tail) = 1 - 0.7^n$$

## Set Theorem $\longleftrightarrow$ Probability Theory

$$A \cap B \longleftrightarrow A \text{ and } B$$

$$A \cup B \longleftrightarrow A \text{ or } B$$

$$A^c \longleftrightarrow A \text{ does not happen}$$

$$A \setminus B \longleftrightarrow A \text{ happens } B \text{ does not happen}$$

**Theorem 1.0.5** (Total Probability Theorem).  $B_1, B_2, \dots, B_n$  with  $\forall \mathbb{P}(B_i) > 0$ , partitions  $\Omega$

$$\begin{aligned}\mathbb{P}(A) &= \mathbb{P}((A \cap B_1) \cup \dots \cup (A \cap B_n)) \\ &= \sum_{i=1}^n \mathbb{P}(A \cap B_i) \\ &= \sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)\end{aligned}$$

**Example 1.0.6.** 3 colors of products: 100 blue, 70 red, 30 purple

defective rate: 3% 9% 7%

partition  $\Omega = B \cup R \cup P$

$$\mathbb{P}(B) = \frac{100}{200}, \mathbb{P}(R) = \frac{70}{200}, \mathbb{P}(P) = \frac{30}{200}$$

$$\mathbb{P}(D) = \mathbb{P}(D \mid B) \mathbb{P}(B) + \mathbb{P}(D \mid R) \mathbb{P}(R) + \mathbb{P}(D \mid P) \mathbb{P}(P)$$

$$\mathbb{P}(B \mid D) = \frac{\mathbb{P}(B \cap D)}{\mathbb{P}(D)} = \frac{\mathbb{P}(D \mid B) \mathbb{P}(B)}{\mathbb{P}(D)}$$

## Condition Probability: rejection sampling (naive ver.)

Choose at random

- If defective
  - Make a histogram
- If not defective
  - Start again.

**Example 1.0.7.** A man possesses five coins, two of which are double-headed, one is double tailed, and two are normal. He shuts his eyes, picks a coin at random, and tosses it. What is the probability that the lower face of the coin is a head?

$$\mathbb{P}(DH) = 2/5, \mathbb{P}(DT) = 1/5, \mathbb{P}(N) = 2/5$$

$$\mathbb{P}(LH) = \mathbb{P}(LH \mid DH) \mathbb{P}(DH) + \mathbb{P}(LH \mid N) \mathbb{P}(N)$$

$$\mathbb{P}(LH) = \mathbb{P}(DH) + 1/2 \mathbb{P}(N) = 2/5 + 1/5 = 3/5$$

He opens his eyes and see that the coin is showing heads; what is the probability that the lower face is a head?

$$\mathbb{P}(LH \mid UH) = \frac{\mathbb{P}(LH \cap UH)}{\mathbb{P}(UH)} = \frac{\mathbb{P}(DH)}{\mathbb{P}(DH) + 1/2 \mathbb{P}(N)}$$

He shuts his eyes again, and tosses the coin again. What is the probability that the lower face is a head?

$$\mathbb{P}(LH' | UH) = \frac{\mathbb{P}(LH' \cap UH)}{\mathbb{P}(UH)} = \frac{\mathbb{P}(LH' \cap UH)\mathbb{P}(DH) + \mathbb{P}(LH' \cap UH|N)\mathbb{P}(N)}{\mathbb{P}(UH)}$$

He opens his eyes and see that the coin is showing heads; what is the probability that the lower face is a head?

$$\mathbb{P}(LH' | UH \cap UH') = \frac{\mathbb{P}(LH' \cap (UH \cap UH'))}{\mathbb{P}(UH \cap UH')}$$

He discards this coin, picks another at random, and tosses it. What is the probability that it shows heads?

**Example 1.0.8.** A random  $N$  of dice is thrown. Let  $A_i$  be the even that  $N = i$ , and assume that  $\mathbb{P}(A_i) = 2^{-i}, i \geq 1$ . Thus sume of the scores is  $S$ . Find the probability that:

a.  $N = 2$  given that  $S = 4$ .  $\mathbb{P}(N = 2 | S = 4) = \frac{\mathbb{P}(N=2 \cap S=4)}{\mathbb{P}(S=4)} = \frac{\mathbb{P}(S=4|N=2)\mathbb{P}(N=2)}{\mathbb{P}(S=4|N=1)\mathbb{P}(N=1) + \dots + \mathbb{P}(S=4|N=4)\mathbb{P}(N=4)} =$

b. The largest number shown by an dice is  $r$ , where  $S$  is unknown

$$\mathbb{P}(M_r) = \mathbb{P}(M_r | N = 1)\mathbb{P}(N = 1) + \mathbb{P}(M_r | N = 2)\mathbb{P}(N = 2) + \dots$$

$$\mathbb{P}(M_r | N = n) = \frac{r^n - (r-1)^n}{6^n}$$

4! choices for first row for the first column there are 3! ways to finish it for the second row there are 3 spots remaining. but two have been picked for each spot. so there is 2! choices for two spots put them one for the reaming then 1 way for the remaining

$$4! \times 3! \times 4$$